Gaurav Agarwal

SYSTEMS SOFTWARE · EMBEDDED SYSTEMS · AVIONICS

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Summary

I have 8+ years of experience spanning all phases of the software development lifecycle, with a proven track record of delivering results in diverse environments including cutting-edge aerospace R&D labs, agile startups, and large-scale enterprises. What drives me is to deepen my expertise, embrace state-of-the-art technologies, and contribute to forward-thinking organizations by developing technology-agnostic, innovative solutions in challenging and dynamic domains.

Work Experience

Boeing India Pvt. Ltd. — *Bengaluru, India*

SOFTWARE ENGINEER 3 | PLATFORM SOFTWARE | EMBEDDED & AVIONICS SYSTEMS — *Nov 2019 – Present*

- Delivered **platform software** across **Cockpit Displays, Common Core, Compute Platform, and Cabin Experience systems** on heterogeneous hardware, building **Linux** and **RTOS**-based real-time services with strong focus on reliability, observability, and cross-system integration.
- Designed **scalable backend architectures**, service interfaces, and data models, translating complex system requirements into **modular APIs** and production-ready services while collaborating across teams to drive standards, maintainability, and design consistency.
- Built **containerized services** and automation workflows on Linux platforms, leading **end-to-end development** from feasibility and architecture through implementation and code reviews, ensuring traceable, testable, and maintainable software delivery.
- Developed **middleware, communication frameworks**, and diagnostic tooling, optimizing performance, **fault tolerance**, and data handling across **distributed environments** and hardware-in-loop test systems.
- Implemented **CI/CD pipelines** using **GitLab CI** and **Docker**, automating build, integration, and deployment workflows to improve release velocity, reproducibility, and overall system stability across multiple product lines.
- Configured custom hardware architectures and optimized **Board Support Packages**, including **U-Boot, kernel, Yocto** recipe-based root file systems, cross-compilation toolchains, and device trees, to enable efficient **secure boot** and enhance runtime performance for custom hardware platforms.

Team Indus (Axiom Research Labs Pvt. Ltd.) — *Bengaluru, India*

FLIGHT SOFTWARE ENGINEER | INTEGRATED AVIONICS | COMMAND & DATA HANDLING | GNC — *Jan 2017 – Oct 2019*

- Developed software systems for **orbital, descent and surface** phases of the **soft landing lunar mission**, with onboard state estimation, autonomous attitude correction, lunar terrain feature tracking, active thermal and power control, interface drivers for sensors and peripherals, with limited **fault detection, isolation, and recovery (FDIR)**.
- Developed software system to perform an **autonomous lunar descent sequence**, with the onboard estimation of lander states, constrained landing site selection, targeted descent to the selected landing site and mode transition logic.
- Design and development of **onboard data handling telemetry & telecommand** packet definition for the entire lunar landing mission: real-time, absolute time-tagged, patch, differential time-tagged, configurable block and event-based.
- Developed frameworks for running **regression unit, interface and integration level testing** with auto code generation capabilities involving sensor and interface card emulation using **Interface Emulation Card (IEC)**, board bring-ups for **Integrated Avionics Unit (IAU)**, and automated report generation.
- Built **Processor in Loop Simulation (PiLS)** framework emulating sensor and actuator electrical interfaces to IAU.

Academic Projects

PISAT – Nano-satellite launched aboard PSLV C-35 (26 Sept 2016) *URSC, ISTRAC, Bengaluru, India*

STUDENT TEAM LEAD *Oct 2014 – Dec 2016*

- Complete design, development, assembly, integration and testing of **PISAT** – a nano-satellite student project funded by **ISRO** and **PES University**, executed under the expertise of ex-ISRO scientists.
- **System Engineering**: Subsystem-level requirements collation, design and development lifecycle, complete verification and validation for both hardware and software.
- **OBC & ADCS**: Built real-time software for an imaging satellite in a component-based manner – attitude determination, control systems, telemetry and telecommand on an Atmel microprocessor with bare-metal architecture. Built test frameworks for scenario-based testing, open-loop and closed-loop simulations.
- **Payload**: Developed NanoCam C1U functionality, operations and test bench for complete analysis of camera parameter settings.
- **AIT**: Built robust test system emulating sensors, interface cards and ground software – used for avionics bring-ups, On-board in Loop Simulation (OiLS), independent verification of telemetry, telecommand, payload interface and ground checkout.
- **Mission Planning**: Detailed design documents for CDR, PSR, PLR, sequence of events, in-orbit tracking and post-data analysis.

Technical Skills

Languages	C, C++, Python, Java, Bash, SQL, MATLAB & Simulink, LaTeX
Systems & Platforms	Linux (Ubuntu, CentOS, Yocto, Buildroot), RTOS (DEOS, VxWorks, FreeRTOS), Windows
Cloud & Infrastructure	AWS (EC2, S3, Lambda), Docker, Kubernetes, Nginx, Kafka, Terraform
DevOps & Tools	Git, GitLab CI/CD, Jenkins, CMake, Doxygen, Polyspace, Postman, Vagrant
Protocols	UART, SPI, I2C, CAN, ARINC, TCP/IP, HTTP/REST, MQTT, gRPC
Backend & Data	PostgreSQL, MongoDB, GraphDB, REST APIs, React.js, OAuth, JWT
Architectures	ARM (Cortex-M, Cortex-A), SPARC, RISC-V, Xilinx Zynq (PS-PL)

Education

P.E.S. Institute of Technology (Autonomous, VTU) *Bengaluru, India*

B.E. IN ELECTRICAL AND ELECTRONICS ENGINEERING *Aug 2013 – May 2017*

- GPA: 8.93 / 10.00
- Major Courses: Embedded Systems, Control Systems, Digital Signal Processing

Kerala Samajam Model School (KSMS) *Jamshedpur, India*

I.C.S.E. & I.S.C. – PURE SCIENCE WITH COMPUTER APPLICATION *Mar 1999 – May 2013*

- ICSE: 93.4%, ISC: 88.75%